28 A proton has a kinetic energy of 1.00 MeV. If its momentum is measured with an uncertainty of 1.00 %, what is the minimum uncertainty in its position?

A 4.14×10^{-19} m

B 1.15×10^{-18} m

C 2.87×10^{-12} m

D 1.99×10^{-11} m

Ans: C

KE, $E = \dfrac{p^2}{2m}$

$p = \sqrt{2mE}$

$$p = \sqrt{2(1.67 \times 10^{-27})(1.00 \times 10^{6})(1.6 \times 10^{-19})} = 2.3117 \times 10^{-20} \text{ kg m s}^{-1}$$

$$\frac{\Delta p}{p} = \frac{1}{100}$$

$$\Delta p = (0.01)(2.3117 \times 10^{-20}) = 2.3117 \times 10^{-22} \text{ kg m s}^{-1}$$

$$\Delta x \Delta p \geq h$$

$$\Delta x \geq \frac{h}{\Delta p}$$

$$\Delta x \geq \frac{6.63 \times 10^{-34}}{2.3117 \times 10^{-22}} = 2.87 \times 10^{-12} \text{ m}$$